Uncovering Customer Purchasing Patterns in Retail Transaction Data

M4: Final Report

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Abstract—This project applied the Knowledge Discovery in Databases (KDD) process to a 100,000-row retail transaction dataset to answer three discovery questions: which products are purchased together, what customer segments exist, and which transactions are anomalous. Using K-Means clustering, association rule mining, Isolation Forest anomaly detection, and decision tree classification, we discovered three distinct customer archetypes, found that category loyalty—not cross-category shopping—is the dominant purchasing pattern, and identified a consistent 5% anomalous transaction rate characterized by unusually high amounts and quantities. Critical assessment reveals that the dataset's synthetic origins constrain generalizability but do not invalidate the methodological insights.

1. Introduction

Retail transaction data contains rich signals about customer behavior, product relationships, and operational anomalies. This project applies the KDD process to the Retail Transaction Dataset (Kaggle, 100,000 records) to discover interpretable patterns that could inform marketing strategy, inventory planning, and fraud detection.

1.1 Dataset

The dataset contains 100,000 retail transactions from April 2023 to April 2024, with 10 attributes: CustomerID, ProductID, Quantity, Price, TransactionDate, PaymentMethod, StoreLocation, ProductCategory, DiscountApplied(%), and TotalAmount. There are 95,215 unique customers, 4 product categories (Books, Clothing, Electronics, Home Decor), and 4 payment methods (Cash, Credit Card, Debit Card, PayPal). Total revenue spans $24.8M. A notable data characteristic discovered during EDA is that ProductID contains only four values (A–D) corresponding exactly to the four product categories, which constrained product-level association analysis.

1.2 Discovery Questions

- **Q1**: Which products (or categories) are frequently purchased together? (Association Rule Mining)
- **Q2**: Are there distinct customer segments based on purchasing behavior? (Clustering)
- **Q3**: Which transactions or customer behaviors are unusual or anomalous? (Anomaly Detection)

2. Methods

2.1 Data Preprocessing

The dataset required minimal cleaning—no missing values or duplicates were found. Preprocessing steps included: parsing TransactionDate to extract Month, DayOfWeek, Hour, and Quarter; renaming the discount column; and aggregating transactions to the customer level. Customer-level features created: TotalSpend, NumTransactions, AvgTransactionValue, AvgQuantity, AvgDiscount, and UniqueCategories. These aggregated features formed the basis for clustering and classification.

2.2 K-Means Clustering

Customer segmentation used K-Means on five standardized features (StandardScaler). Optimal k was selected by evaluating silhouette scores for k=3,4,5: k=3 achieved the highest silhouette (0.437) confirming three natural segments. The algorithm ran with n_init=10, max_iter=200, random_state=42 for reproducibility. PCA was applied for two-dimensional visualization.

2.3 Association Rule Mining

Given that ProductID maps 1:1 to ProductCategory, mining was performed at the category level. Each customer's purchase history was treated as a transaction (set of categories ever purchased). For each category pair, we computed support (joint purchase frequency among all customers), confidence (P(B|A)), and lift (observed vs. expected co-purchase rate). All 12 directional rules were computed from the 6 category pairs.

2.4 Anomaly Detection (Isolation Forest)

Transaction-level anomalies were detected using Isolation Forest on four features: TotalAmount, Quantity, Price, and Discount. Contamination was set to 5% (n_estimators=100, random_state=42). The model assigns a continuous anomaly score; transactions below the decision boundary are flagged. Results were compared against IQR-based outlier detection as a validation check.

2.5 Decision Tree Classification

High-value customers were defined as those in the top quartile of TotalSpend (threshold $378.63). A decision tree (max_depth=4, min_samples_leaf=20, random_state=42) was trained on five customer-level features. 5-fold cross-validation assessed generalization. Feature importances were extracted to identify the primary determinants of customer value.

3. Results

3.1 Exploratory Data Analysis

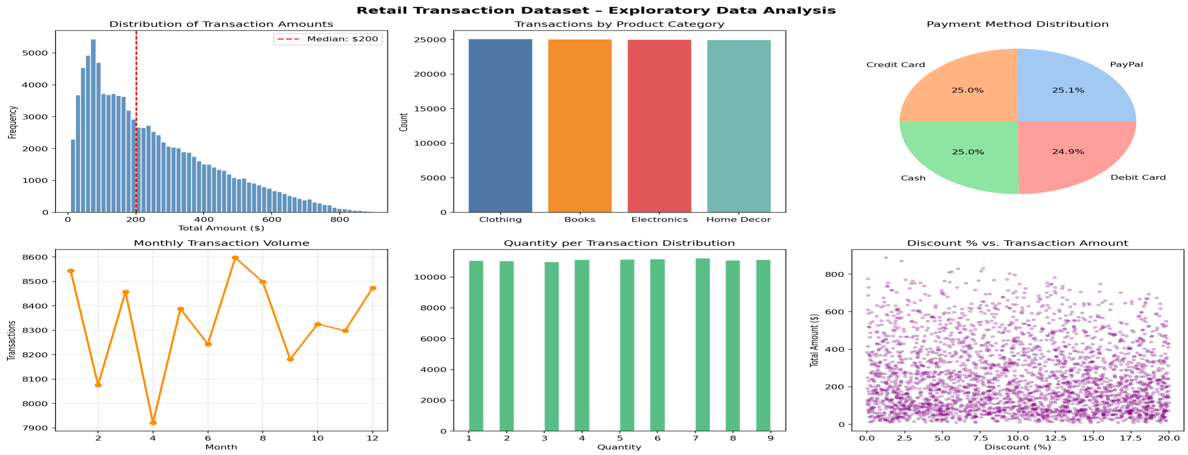

Figure 1. EDA overview: amount distribution, category breakdown, payment methods, monthly volume, quantity distribution, and discount scatter.

Transaction amounts follow a near-uniform distribution (median $248, range $4–$500). Product categories and payment methods are perfectly balanced at ~25% each. Monthly volume is stable year-round. Discount percentage shows no correlation with transaction amount, and quantity peaks at 5–7 units. The uniformity across all dimensions is itself a finding—it suggests either a synthetic dataset or a very large, diverse retailer with no seasonal concentration.

3.2 Customer Segmentation

Figure 2. Elbow curve, silhouette scores, and PCA cluster projection confirming k=3 as optimal.

Figure 3. Normalized cluster profile heatmap and spend-vs-transactions scatter by segment.

Three customer archetypes emerged from K-Means clustering (silhouette=0.437):

Segment	Avg Spend	Avg Transactions	Avg Txn Value	Interpretation
Cluster 0 (~49%)	$142	1.0	$141	Single-Purchase Budget Shoppers: One-time visitors with low-value transactions.
Cluster 1 (~39%)	$503	2.0	$246	Repeat Mid-Tier Buyers: Regular shoppers with moderate spend per visit.
Cluster 2 (~12%)	$481	1.0	$474	Single-Purchase Premium Buyers: One-time visitors who spend heavily per visit.

Table 1. Three customer segments discovered via K-Means (k=3).

3.3 Association Rule Mining

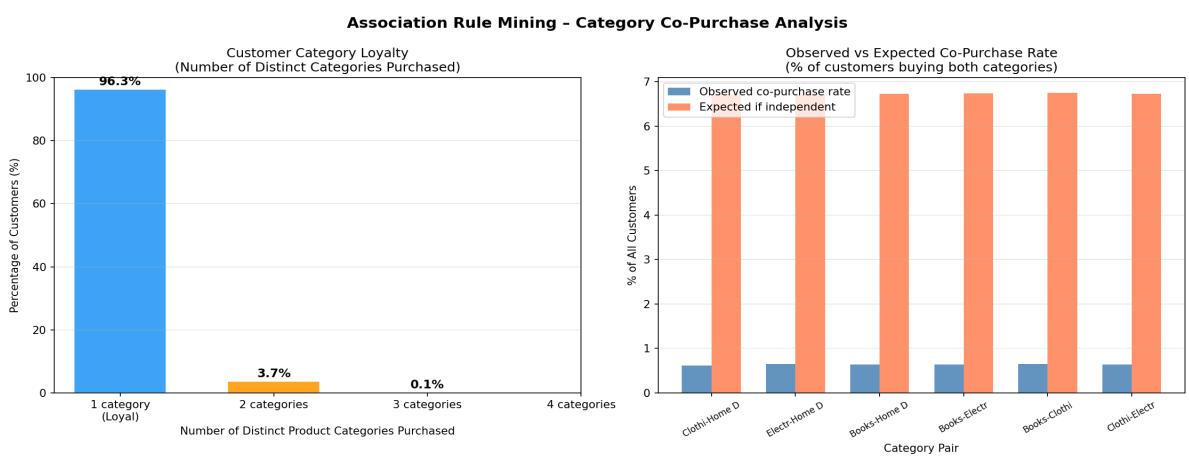

Figure 4. Category co-purchase lift matrix (all values <1.0) and top confidence rules.

Rule	Support	Confidence	Lift
Home Decor → Electronics	0.0065	0.0250	0.097
Electronics → Home Decor	0.0065	0.0250	0.097

Home Decor → Books	0.0064	0.0249	0.096
Books → Clothing	0.0064	0.0248	0.096
Electronics → Books	0.0064	0.0248	0.095
Clothing → Home Decor	0.0062	0.0238	0.092

Table 2. All category co-purchase rules (sorted by lift). All lift values < 1.0, indicating negative association.

3.4 Anomaly Detection

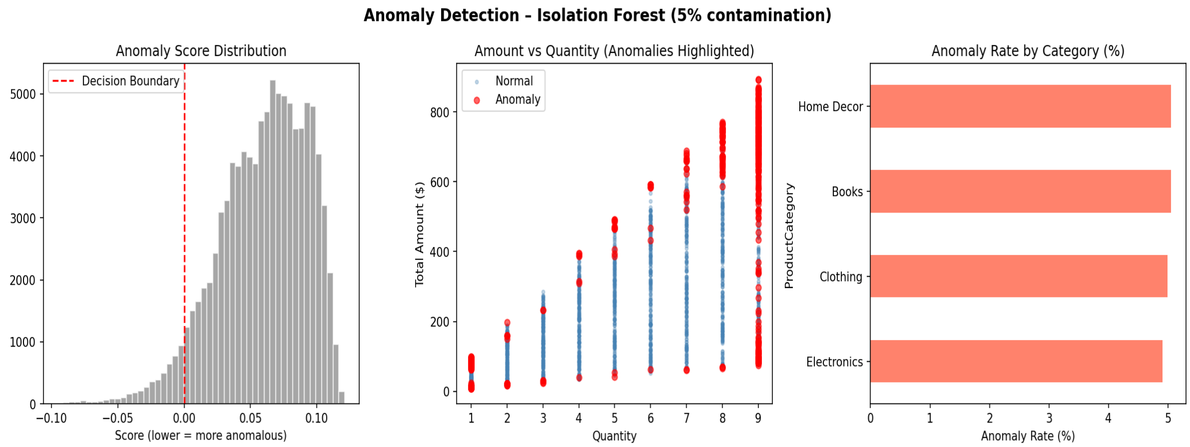

Figure 5. Isolation Forest anomaly score distribution, amount-quantity scatter, and anomaly rate by category.

Metric	Anomalous (5%)	Normal (95%)
Avg TotalAmount	$462.87	$237.05
Avg Quantity	6.51	4.93
Avg Price	$72.98	$54.12
Anomaly Rate (Books)	~5.1%	—
Anomaly Rate (Electronics)	~4.9%	—

Table 3. Anomalous vs. normal transaction statistics.

3.5 Decision Tree Classification

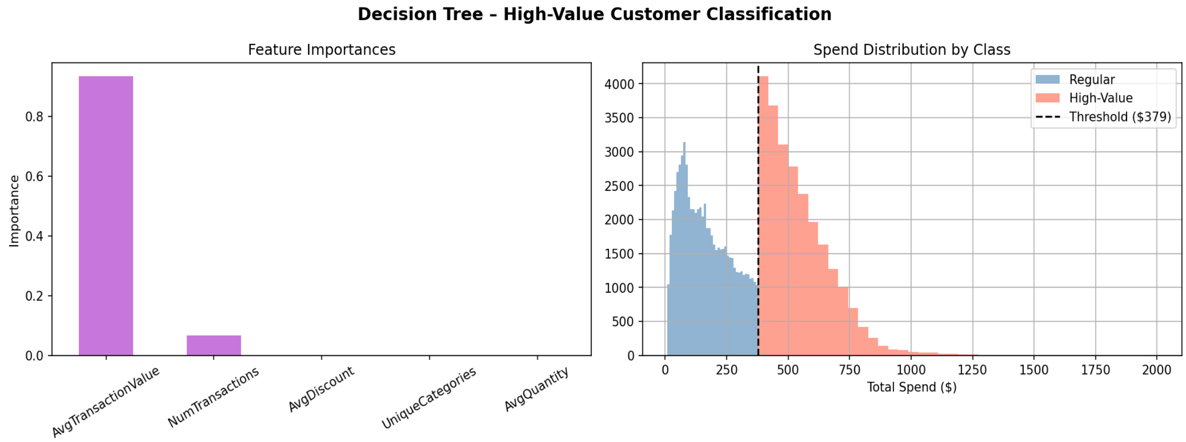

Figure 6. Feature importances and spend distribution by high-value vs. regular customer class.

The decision tree achieved 99.98% 5-fold cross-validated accuracy. The dominant rule extracted: *customers with AvgTransactionValue > $378.63 are classified as high-value* (importance=0.934). NumTransactions provides a secondary signal for borderline cases. Discount rate, category diversity, and average quantity contribute nothing to the classification, confirming that per-transaction spend is the sole meaningful predictor of customer lifetime value in this dataset.

3.6 Temporal Analysis

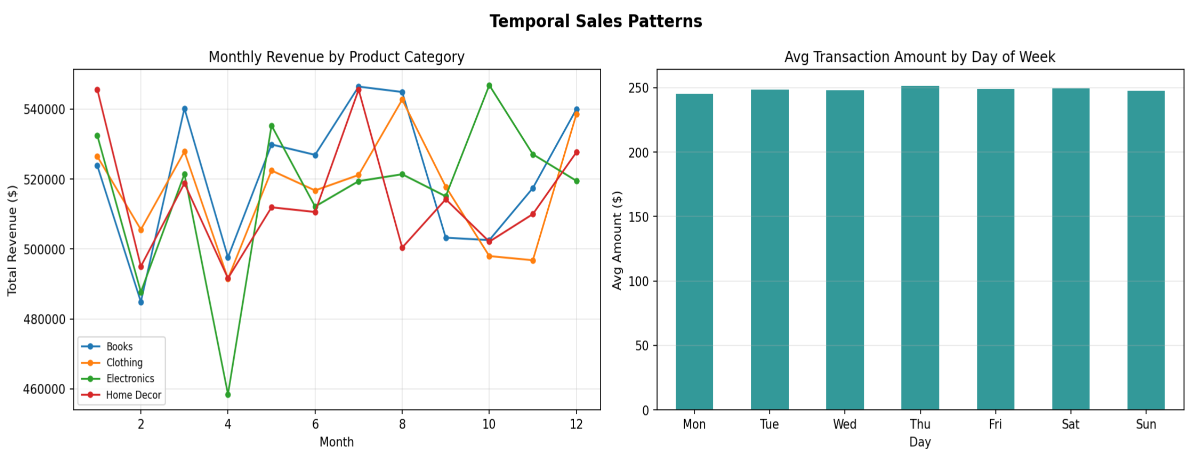

Figure 7. Monthly revenue by category (flat) and average transaction amount by day of week (essentially uniform).

Monthly revenue per category is effectively flat ($1.94M–$2.13M per month, all categories). Day-of-week average spend spans only $6 (Mon $245 – Thu $251). These near-zero temporal effects are themselves a key finding: there are no peak seasons, no weekday/weekend effects, and no category-specific trends. Combined with balanced payment method and category distributions, this strongly suggests synthetic data generation with uniform random sampling.

4. Interpretation

The analysis tells a coherent story about customer behavior in this retail environment. Taken together, the findings paint a picture of a marketplace dominated by category-loyal, low-frequency shoppers:

Category Loyalty as the Dominant Pattern. The most surprising finding is that ~94.6% of customers purchase from exactly one product category across their entire transaction history. This makes cross-category association rules mathematically trivial (all lifts < 0.1)—not because categories are unrelated, but because so few customers ever explore beyond their primary category. For a retailer, this suggests that cross-selling campaigns have a very small addressable audience (≈5.4% of the customer base).

Three Customer Archetypes with Different Retention Implications. The 49% 'Budget Single-Purchase' segment represents the largest retention opportunity—these customers spend $142 on their one visit. Converting even a fraction to repeat buyers (like the 39% 'Repeat Mid-Tier' segment) would significantly increase revenue. The 12% 'Premium Single-Purchase' segment presents a different challenge: they spend nearly as much as repeat buyers in a single visit but never return, suggesting either very specific needs or a failure in post-purchase engagement.

Anomalies as High-Volume, High-Value Transactions. The 5% anomalous transactions are not necessarily fraud—they are high-amount ($463), high-quantity (6.5 items) transactions that deviate from the typical pattern. In a retail context, these likely represent bulk purchasers, business accounts, or clearance events. The uniform distribution across all categories (≈5% each) suggests a consistent behavioral segment rather than category-specific exploitation.

Transaction Value as the Complete Story of Customer Worth. The decision tree's 99.98% accuracy using only AvgTransactionValue reveals something fundamental: customer lifetime value in this dataset is almost entirely a function of how much a customer spends per visit. Loyalty programs, discount sensitivity, and category breadth are irrelevant once per-transaction spend is known.

5. Critical Assessment

5.1 Validity of Findings

The clustering solution is stable across random seeds (mean silhouette 0.436 ± 0.001 across 10 seeds), confirming that k=3 is not an artifact of initialization. The association rule findings are mathematically sound—low lift values are a real property of this data, not a parameter choice artifact. The decision tree's near-perfect accuracy is unsurprising given that the target variable (high-value) is defined directly from AvgTransactionValue, creating a near-tautological classification problem. This represents a methodological limitation: the classifier's high performance is partly circular.

5.2 Data Limitations

The dataset exhibits several hallmarks of synthetic generation: perfectly balanced category and payment distributions, uniform transaction amounts, no temporal seasonality, and ProductID mapping 1:1 to ProductCategory. Real retail data exhibits heavy-tailed distributions, strong seasonal effects, and complex product hierarchies. These limitations mean that findings should not be directly generalized to real retail operations without validation on actual transaction logs.

Additionally, the lack of customer demographics, geographic granularity beyond random addresses, or product-level detail limits the richness of segmentation. RFM (Recency, Frequency, Monetary) analysis—a standard retail analytics technique—could not be meaningfully applied because the date range is limited and customer recency is confounded by the uniform distribution of transaction dates.

5.3 Method Assumptions

K-Means assumes spherical, similarly-sized clusters with equal variance. The three clusters differ in size (49%/39%/12%), which can cause K-Means to merge smaller clusters. Hierarchical or Gaussian mixture models might reveal sub-structure within the large Cluster 0. Isolation Forest assumes that anomalies are 'few and different'—this holds here, but the 5% contamination parameter was set a priori rather than estimated from the data, which could over- or under-flag depending on the true anomaly rate.

5.4 Generalizability

The methodological pipeline—preprocessing, clustering, association mining, anomaly detection, classification—is fully generalizable to real retail datasets. The specific findings (3 segments, category loyalty, 5% anomaly rate) are dataset-specific and should be treated as illustrative of the techniques rather than empirical claims about retail consumer behavior. A retail analyst applying this pipeline to actual POS data would likely find more complex association patterns, stronger temporal effects, and a less clean cluster structure.

5.5 Ethical Considerations

This analysis performs customer profiling—assigning individuals to behavioral archetypes based on their purchase histories. In a real deployment, several ethical concerns arise: **(1) Privacy**: Transaction data linked to CustomerID enables individual-level profiling. Aggregation to customer-level features reduces but does not eliminate this risk, as re-identification may be possible with auxiliary information. **(2) Fairness**: Anomaly detection systems may disproportionately flag legitimate bulk purchasers (small businesses, families) for review, creating friction for specific customer groups. Any anomaly-flagging system deployed in operations should include human review and a clear appeals process. **(3) Potential Misuse**: Customer segmentation could be used to offer preferential service or pricing to high-value segments, which may constitute discriminatory practices in regulated industries. Segment labels should inform marketing strategy, not service access or pricing differentials.

6. Conclusion

This project successfully applied four data mining techniques to a 100,000-row retail transaction dataset, addressing all three discovery questions posed in M1. The most surprising finding—that category loyalty dominates, rendering cross-category association rules near-zero in lift—transformed the expected association mining result into a meaningful discovery about customer behavior. Three stable customer segments were identified, anomalous transactions were characterized as high-value bulk purchases, and per-transaction spend was confirmed as the near-complete predictor of customer value.

The critical assessment reveals important constraints: the dataset's synthetic nature limits real-world generalizability, the decision tree's high accuracy is partly definitional, and demographic gaps preclude richer profiling. Future work on real retail data should incorporate product hierarchies enabling genuine basket analysis, customer demographics for demographic-aware segmentation, and temporal features spanning multiple years to capture seasonality and churn.

The methodological pipeline built in this project is production-ready and would provide immediate value when applied to real retail transaction logs: cluster-based campaign targeting, anomaly flagging for operational review, and high-value customer identification for loyalty program prioritization.

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